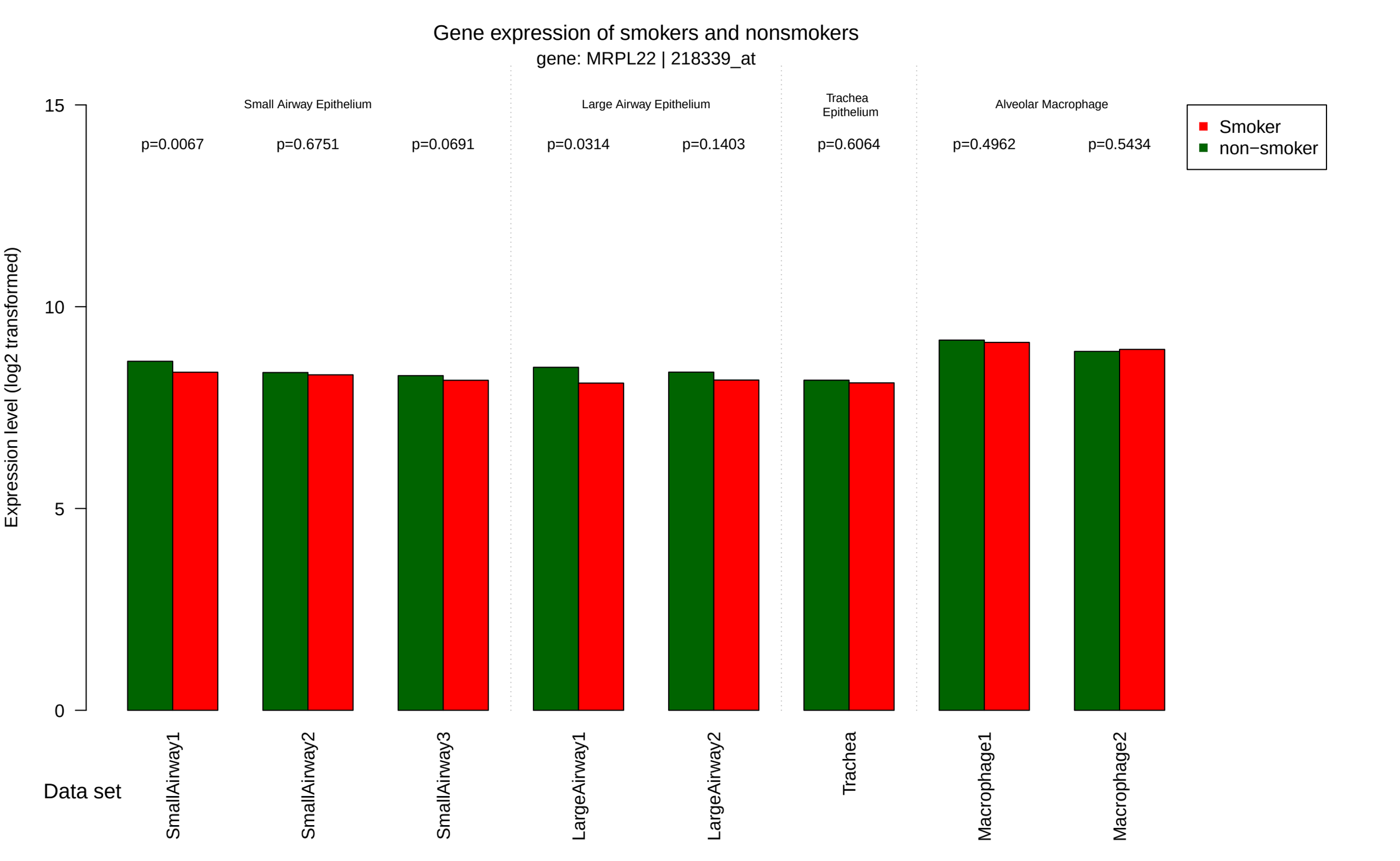
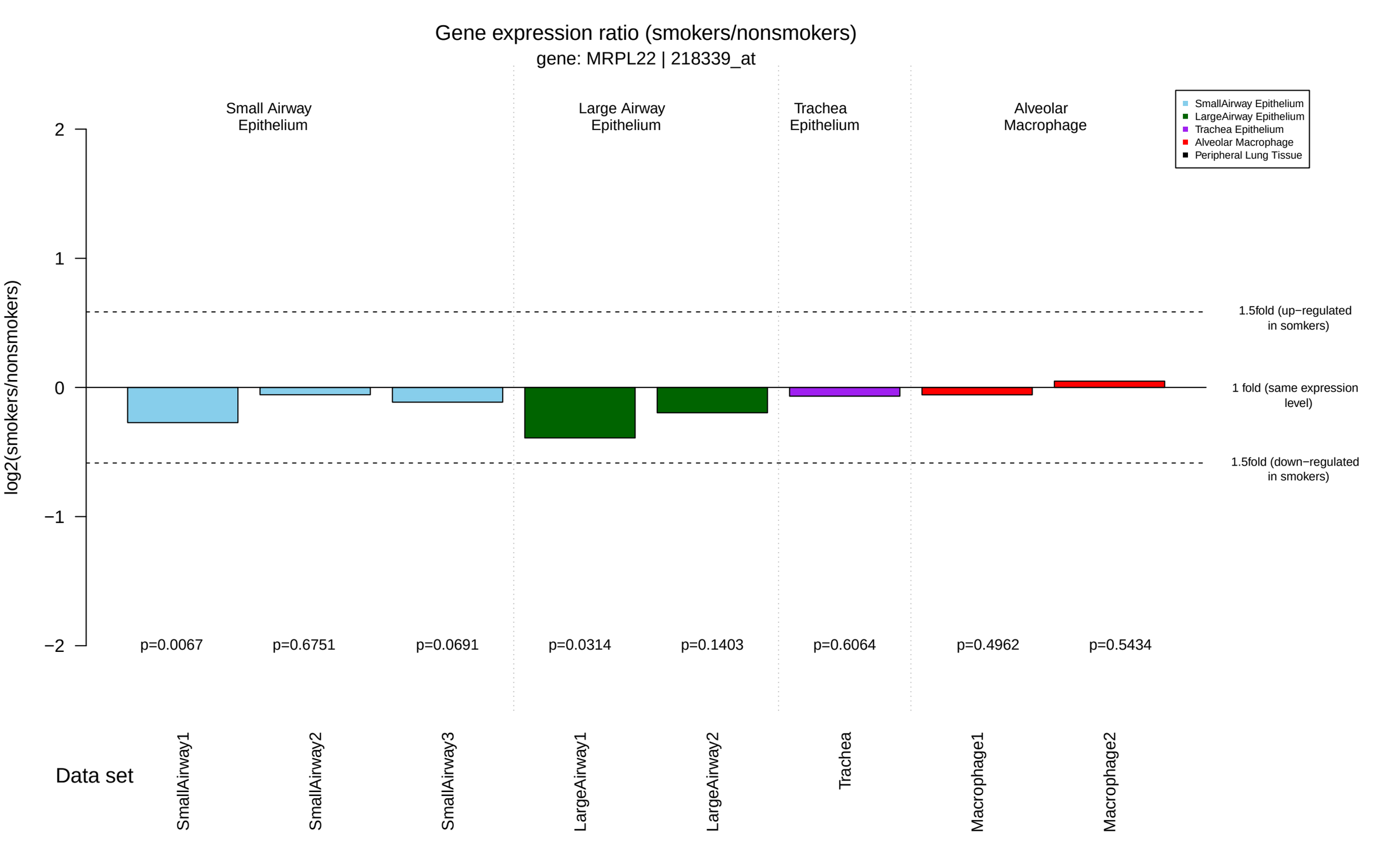
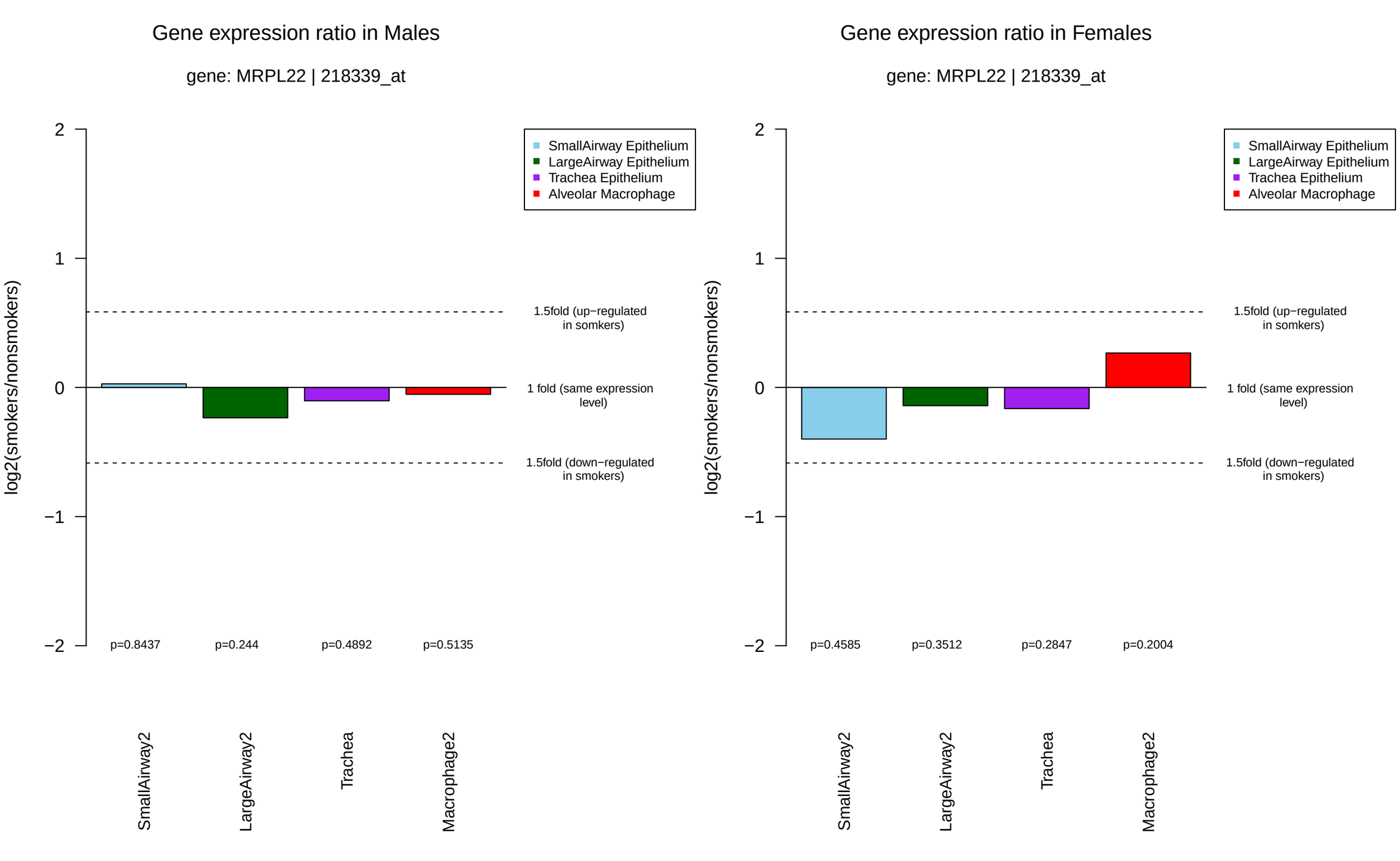
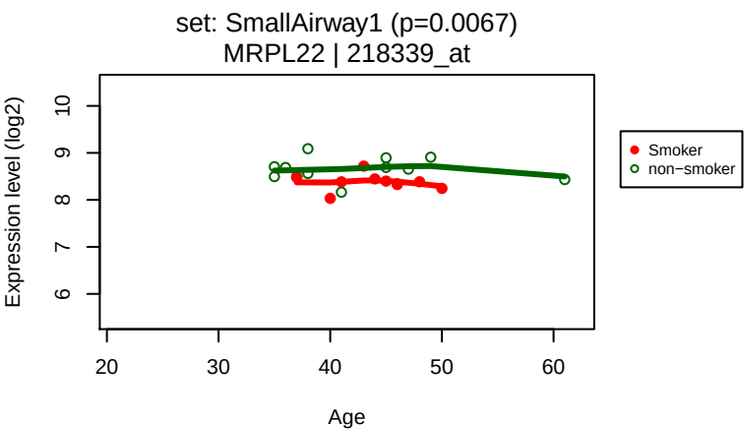
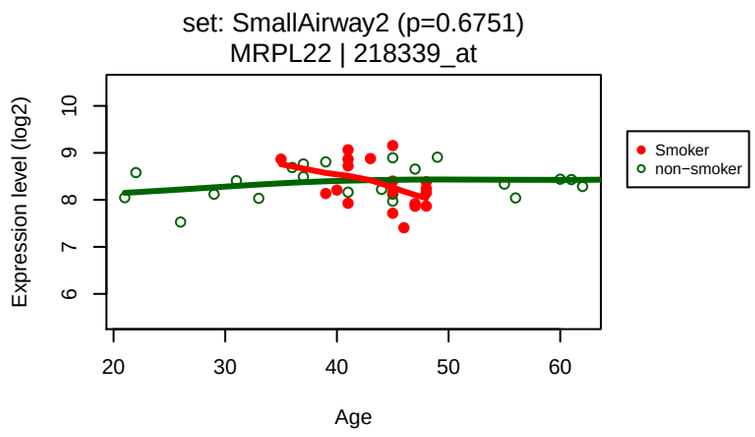
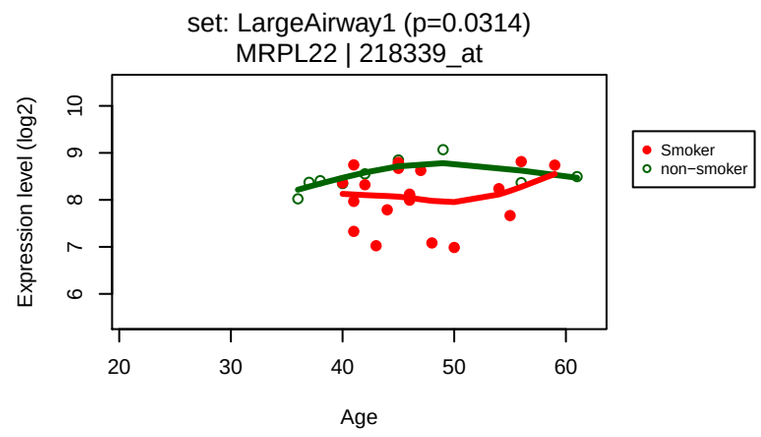

Gene expression of smokers and nonsmokers
gene: MRPL22 | 218339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0067
p=0.6751
p=0.0691
p=0.0314
p=0.1403
p=0.6064
p=0.4962
p=0.5434
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

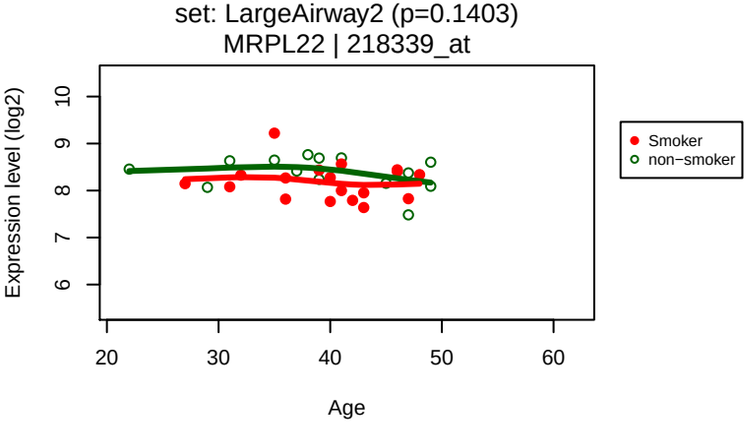

Gene expression ratio (smokers/nonsmokers)
gene: MRPL22 | 218339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0067
p=0.6751
p=0.0691
p=0.0314
p=0.1403
p=0.6064
p=0.4962
p=0.5434
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

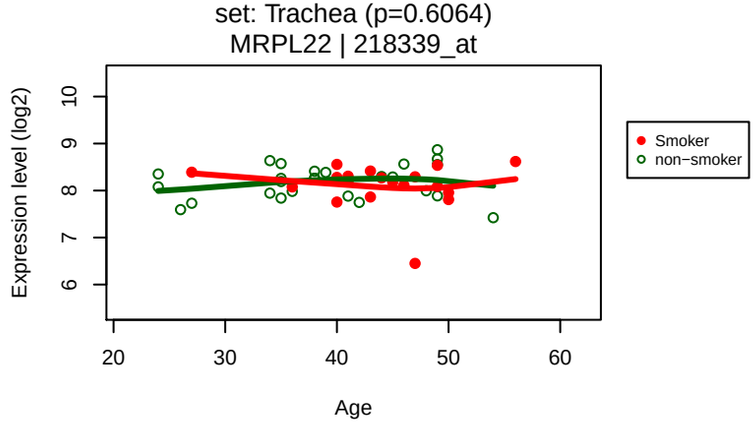

Gene expression ratio in Males
gene: MRPL22 | 218339_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8437
p=0.244
p=0.4892
p=0.5135
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL22 | 218339_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4585
p=0.3512
p=0.2847
p=0.2004
SmallAirway2
LargeAirway2
Trachea
Macrophage2

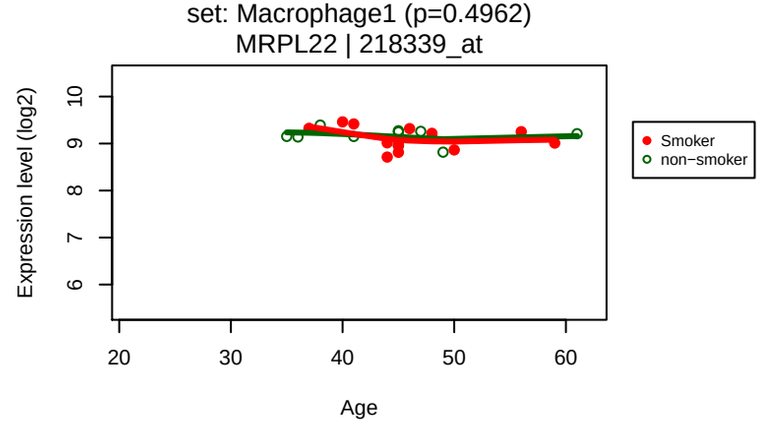

set: SmallAirway1 (p=0.0067)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker

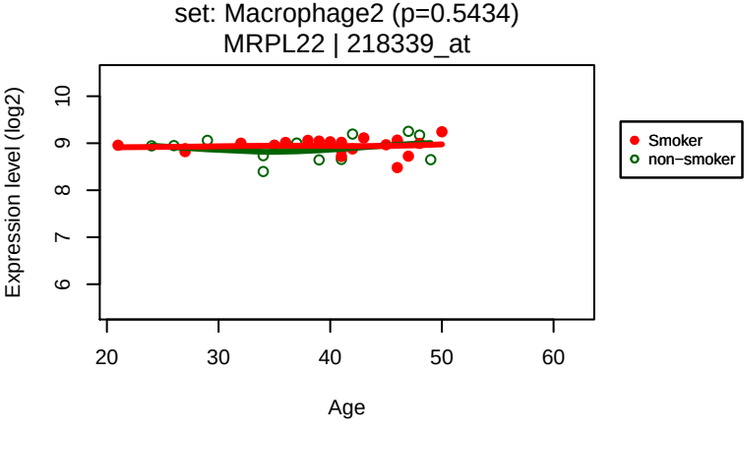

set: SmallAirway2 (p=0.6751)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker

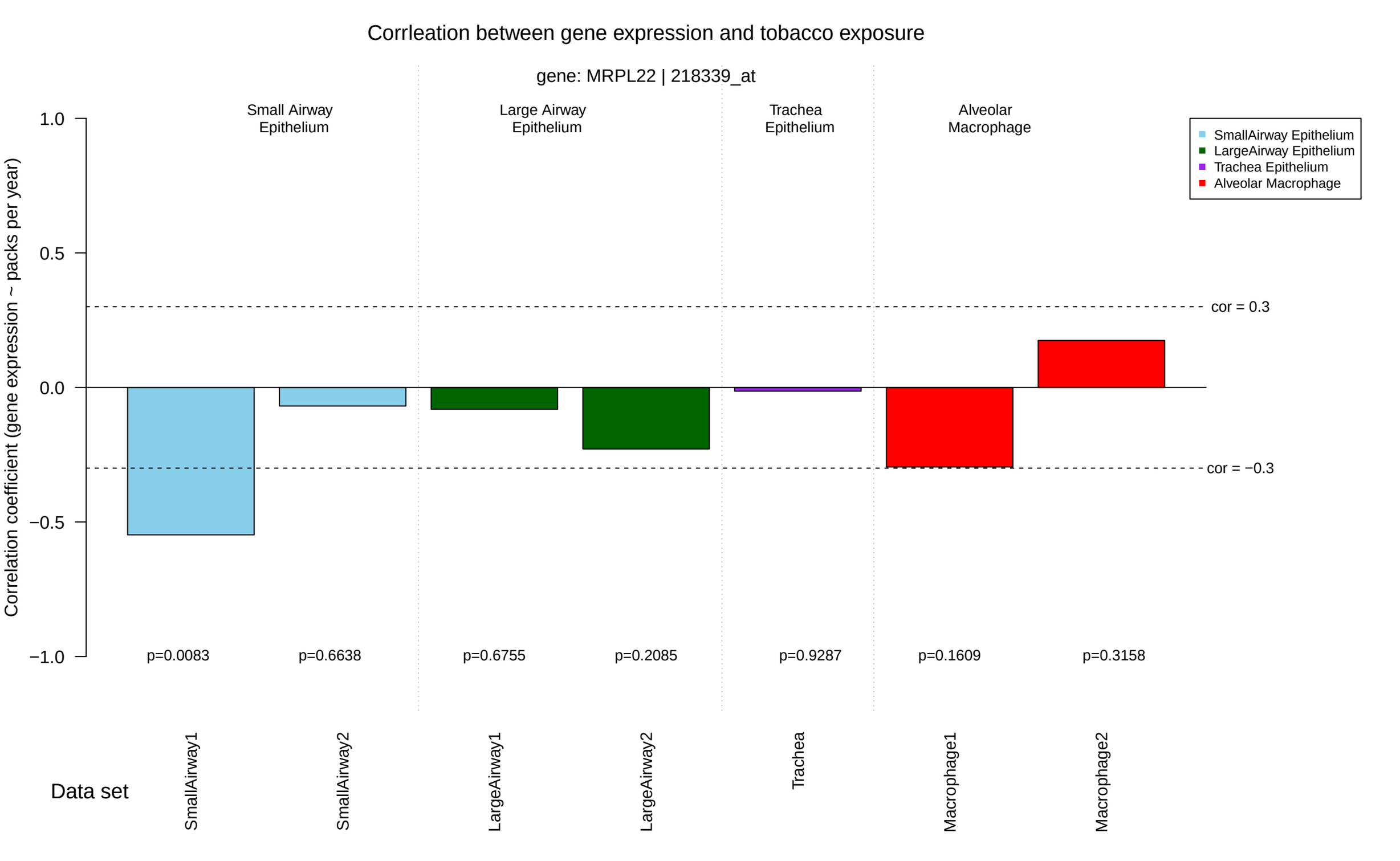

set: LargeAirway1 (p=0.0314)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1403)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6064)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4962)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5434)
MRPL22 | 218339_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL22 | 218339_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0083
p=0.6638
p=0.6755
p=0.2085
p=0.9287
p=0.1609
p=0.3158
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2